The Bagel Game

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> *Some things, the existence of which cannot be proved or considered probable, but which, precisely because pious and conscientious people treat them as something that really exists, come a little nearer to being able to exist and be born.*
>
> \- H. Hesse. The Bead Game

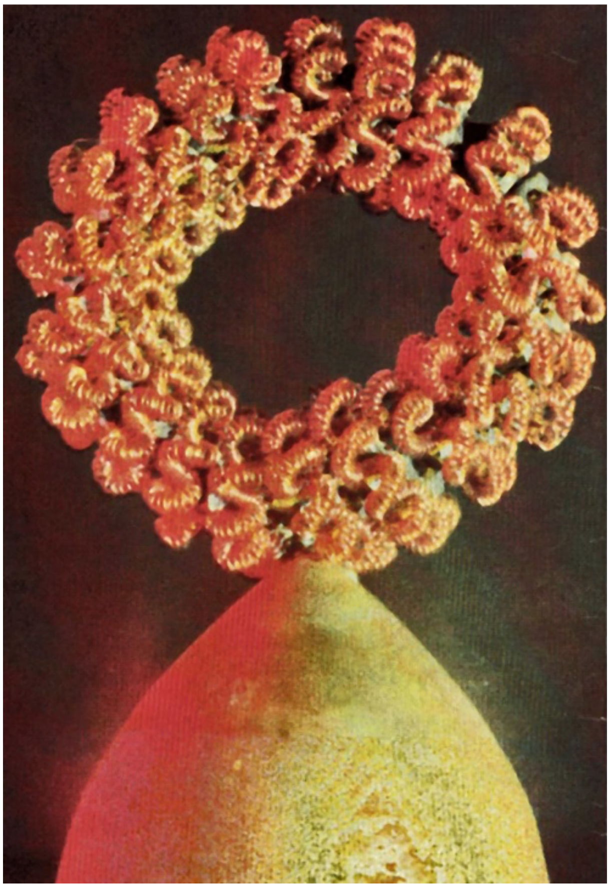
About seven years ago, in a Moscow basement, which housed a mysterious organization, the employees of which did not really know where it came from and where it disappeared to, I saw a real ghost. But the ghost was not in the style of the English one - with chains, sad sighs, etc. - but quite modern, scientifically referred to as an electromagnetic phantom.

The thing was...

The story went like this. On the laboratory table there was an ingenious coil - a wire spiral about 1 cm in diameter, twisted into another spiral about 5 cm in diameter, and then coiled into a bagel-torus about 30 cm in diameter. The coil was connected to a low-frequency generator, and above it, at a height of about half a meter, was fixed a sensor of a device designed to register weak alternating magnetic fields - the so-called microteslameter.

The generator was switched on, and on the screen of the electron-beam tube of the microteslameter there was a bright sinusoid, indicating that the alternating current (with a frequency of about 30, rather than the mains frequency - 50 Hz), flowing through the coil, generates on its axis a variable magnetic field. If the sensor was moved away from the coil axis, the signal was weakened and, therefore, it was indeed created by the magnetic field of the circular current flowing through the coil, and not by any extraneous parasitic induction on the microteslameter.

Then the generator was unplugged, and the coil was unplugged from the generator. What do you expect? The bright sinusoid on the screen of the electron-beam tube of the microteslameter did



not disappear, and in general did not change! And, as before, the signal weakened if the sensor was moved away from the coil axis... Naturally, the first thing I suspected was some elementary cheating. But after checking the circuit, I was convinced that there was no way current could flow through the coil; there were no sources of alternating magnetic field with a non-standard frequency around the coil.

Where is the signal coming from, then? Probably some kind of first-class cheating, I thought. However, it was explained to me that there was an electromagnetic phantom in front of me, i.e., a kind of ghost of an electromagnetic field. No one was able to explain the essence of this phenomenon to me, so I just had to watch and be amazed.

Not only that. The next thing I heard was really fantastic. If you carefully remove the coil from the table, the phantom will not only not disappear, but will remain in the same place, where it was born. And it can exist, gradually weakening, for up to two days. However, I was warned that this experience is not always successful, and the electromagnetic ghost itself turns out to be quite skittish - it can be destroyed by an electric discharge produced nearby, or you can simply sweep it off the table with your hand.

I couldn't take it anymore. I shrugged my shoulders and kept a benevolent smile (though I wanted to twist my finger at my temple), and I retreated without waiting for the end of the demonstration of what could not be, because it could never be.

I still deeply regret that I didn't have a little more patience and curiosity at the time.

Geometry of electromagnetism

I did not believe in the existence of the electromagnetic phantom, although I saw it (if you can say that about an invisible ghost) with my own eyes. However, emitters of toroidal shape have long been of interest to me as possible sources of asymmetric electromagnetic fields: I assumed that such fields would have an increased biological activity (see "Chemistry of Life", 1980, № 12, pp. 81-87), and wanted to check this assumption experimentally.

So, I asked my acquaintance, physicist N.E. Nevesky, Candidate of Physical and Mathematical Sciences, of the Institute of Theoretical Problems of the USSR Academy of Sciences, to calculate what the configuration of electromagnetic fields created by multi-spiral type emitters should be.

There is no point in presenting these calculations here (they are published in The Journal of Electricity, 1993, No. 12, pp. 49-52), and a qualitative picture can be obtained simply from the symmetry of Maxwell's equations describing electromagnetic phenomena (see I.S. Zheludev's book "Symmetry and its Applications", Nuclear Research Institute, 1976, p. 246).

If a current flows through a ring-shaped conductor, then a magnetic field arises around this ring, all the lines of force of which are closed to themselves, have neither beginning nor end and are therefore depicted without arrows (the magnetic field strength $\overline{H}$ is an axial vector); if the current changes its direction with frequency ω, then a variable magnetic field will be observed on the ring axis (Figure 1a), and in its plane electromagnetic radiation with a wave length $\lambda = c/\omega$, where c is the speed of light in the vacuum would propagate in all directions.

Such a ring can be considered the primary element from which toroidal emitters of any complexity can be constructed; therefore, for convenience we will call such a ring a torus of the first order, or 1-torus.

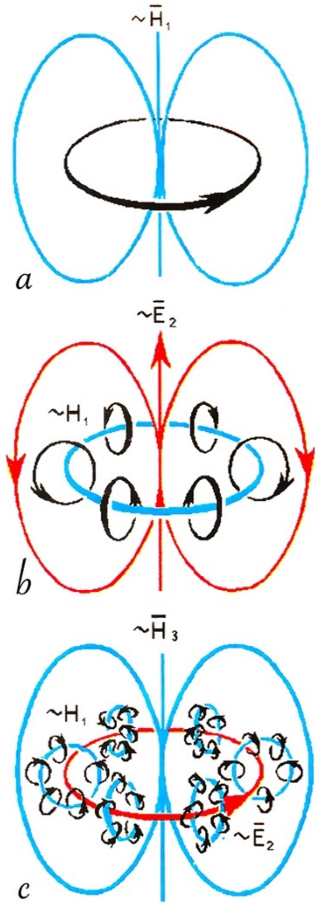


Fig. 1

Electric and magnetic fields of toroidal emitters of different degrees of complexity

a - torus of the first order (1-torus),

b - torus of the second order (2-torus),

c - torus of the third order (3-torus).

Electric field is shown in red, the magnetic one in blue

Identical 1-toruses, i.e., simple conducting rings through which alternating current of the same frequency flows synchronously and in-phase, can be assembled, like from parts of a child's construction set, into a torus of the second order, or a 2-torus. A ring-shaped magnetic field will arise and disappear inside such a real torus, and an alternating electric field with the same frequency, but with a phase shift of $\pm \lambda/2$, will arise and disappear outside it. It is significant that all the lines of force of the external electric field of the 2-torus turn out to be closed in on themselves, like the lines of force of the magnetic field of the 1-torus (Fig. 1b) but depicted by arrows (the electric field strength $\overline{E}$ is a polar vector).

Now, if a torus of the next order, a 3-torus (a supertor, so to speak), is composed of 2-tors, a ring-like magnetic field will arise and disappear inside the 1-tors and will be formed with frequency ω, a ring-like electric field with $\pm \pi/2$ phase shift inside 2-tors, and a magnetic field coinciding in phase with the magnetic field of 1-torus again outside 2-tors (Fig. 1c).

Nevesky paid attention to a very interesting feature of such complex tori: when passing from a torus of one order to a torus of the next order, the external magnetic field $\overline{\mathrm{H}}$ is each time replaced by an electric field $\overline{\mathrm{E}}$ and vice versa, forming the sequence:

$$\overline{\mathrm{H}}_1 \to \overline{\mathrm{E}}_2 \to \overline{\mathrm{H}}_3 \to \cdots \to \overline{\mathrm{H}}_{2n+1} \to \overline{\mathrm{E}}_{2n+2} \to \overline{\mathrm{H}}_{2n+3} \to \cdots,$$

Where n can increase from zero to infinity. And each time the transition from $\overline{\mathrm{H}}$ to $\overline{\mathrm{E}}$ and from $\overline{\mathrm{E}}$ to $\overline{\mathrm{H}}$ can be made by exactly the same kind of substitutions to the equations describing these fields.

In other words, the sequence of uniformly complicating toroidal geometric structures and the configuration of the electromagnetic fields generated by them, form a kind of mathematical recursion, indicating an explicit connection of Maxwell's equations with pure geometry. Namely, since Maxwell's equations are invariant with respect to the Lorentz transformations, which in turn underlie the theory of relativity, the discovered pattern indicates that space-time has the same geometry as the n-tor in which $n \to \infty$.

Such an n-tor has one very interesting feature: any small part of it is indistinguishable from any large part of it. Such self-similar structures (that is, structures in which the part is always equal to the whole) are called fractal and spontaneously arise during processes of very different physical nature - be it a lightning strike or the formation of a tree crown.

The principle of construction of the simplest two-dimensional fractal is shown in Fig. 2; the same way three-dimensional fractals like the channel of linear lightning or the crown of a tree are constructed. But n-tor is, apparently, the only possible three-dimensional fractal which fills space without any residue in such a way that there are no inhomogeneities or isolated branches in it.

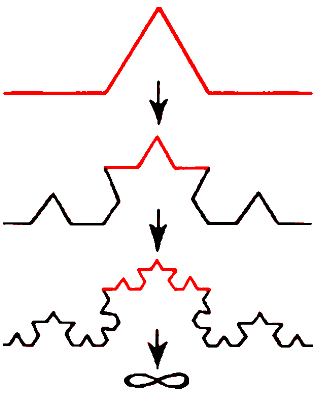


Fig. 2

The principle of constructing the simplest two-dimensional fractal - a structure in which the largest is indistinguishable from the smallest

Maxwell gives the "go-ahead"

All this is, so to speak, pure theory. But practically we can make a toroidal emitter of any order only by twisting it from one whole piece of wire, the ends of which are connected to an alternating current source. Therefore, the electromagnetic field of each next-order torus will be superimposed on the field of the previous-order torus.

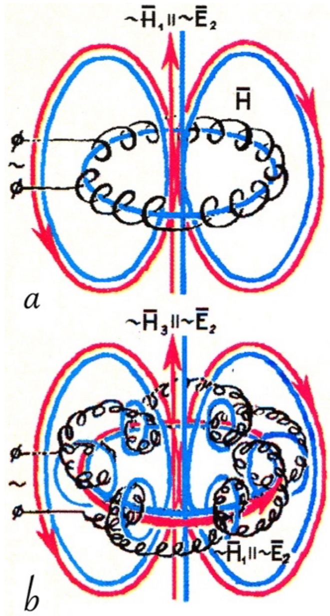


Fig. 3

Electric and magnetic fields of spiral toroidal emitters:

a - spirals of the second order (1,2-torus),

b - spirals of the third order (1,2,3-torus).

In spirals of the third and higher orders, formation of electromagnetic phantoms is possible

Thus, a wire spiral coiled into a 2-tor will create a field combining features of the 1- and 2-tor fields, since the 1-tors forming the 2-tor are connected with each other (Fig. 3a); therefore, we will call such a spiral a 1,2-tor. Accordingly, a 3-tor made of wire will create an electromagnetic field combining features of the fields of 1-, 2-, and 3-tors (Fig. 3b), and we will call it a 1,2,3-tor. And so on.

The coil, which I was shown and in which the ghost of the electromagnetic field arose, just represented a 1,2,3-tor, and in it could arise the variables $\overline{\mathrm{E}}$ and $\overline{\mathrm{H}}$ with closed field lines, shown in Fig. 3b. This solves the mystery of the electromagnetic phantom.

According to the law of electromagnetic induction, a decrease in $\overline{\mathrm{H}}$ is accompanied by an increase in $\overline{\mathrm{E}}$ and vice versa with a phase shift of $\pm\,\pi/2$. In a normal electromagnetic wave, the vectors $\overline{\mathrm{E}}$ and $\overline{\mathrm{H}}$ are perpendicular to each other, and therefore electromagnetic energy is carried along the line perpendicular to the plane in which $\overline{\mathrm{E}}$ and $\overline{\mathrm{H}}$ lie, the flux density of which (that is, the amount of energy carried per unit time through the unit area) is $\overline{P} = \overline{E} \cdot \overline{H}$, where $\overline{P}$ is the polar vector called the Poynting vector (Figure 4a).

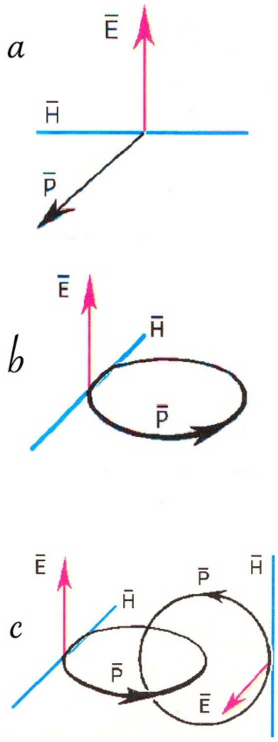


Fig. 4

The mutual orientation of the vectors E, H and $\overline{P} = \overline{E} \cdot \overline{H}$ in spiral toroidal emitters when their size is much smaller than the wavelength $\lambda = c/\omega$:

a - in first- and second-order spirals (1- and 1, 2-tors),

b - in third-order spirals (1, 2, 3-tors),

c - in fourth-order spirals (1, 2, 3, 4-tors).

In 1,2,3-tors, the Poynting vector $\overline{P}$, which describes the electromagnetic energy flux density, closes in a ring, forming a stable phantom. And the electromagnetic phantom formed by 1,2,3,4-tor can serve as a macroscopic model of the structural element of the physical vacuum.

Unlike 1-tor, 1,2-tor can exist in "left" and "right" forms, which differ from each other only as an object and its reflection in a mirror. But 1,2,3-tor is capable of existing in four forms, two of which are completely different because they cannot be combined with each other even by reflection in a mirror.

Indeed, the 1,2,3-tor can be made in two fundamentally different ways: by coiling the left-hand wire helix into a left-hand supertor (or a right-hand helix into a right-hand supertor) or by coiling the left-hand helix into a right-hand supertor (or a right-hand helix into a left-hand supertor). The difference between left-left and left-right (as well as right-right and right-left) supertors can be detected at the very first attempt to make them out of wire: left-right and right-left 1,2,3-tors tend to curl themselves out of left and right 1,2-tors; in contrast, left-left and right-right 1,2,3- tors actively resist their creation by literally tearing themselves out of hands, i.e., striving to give themselves a configuration with a minimum reserve of spare energy.

If a small supertor is connected to a low-frequency generator, then compared to the length of the electromagnetic waves it emits, the emitter itself can be considered almost a point. Thus, the size of the supertor I saw in the basement was relative to the wavelength about the same as the size of a medium-sized molecule to the diameter of a ping-pong ball. So, for such a torus, the laws of electromagnetic induction would remain the same, but depending on whether we are dealing with left-left or left-right (as well as right-right or right-left) supertors, the phase shift between $\overline{E}$ and $\overline{H}$ could be either $+\pi/2$ or $-\pi/2$. And that solves everything.

If the phase shift is such that the $\overline{E}$ and $\overline{H}$ vectors have the same mutual orientation as in an ordinary electromagnetic wave, then the energy supplied to the supertor will dissipate in the usual way. But if their mutual orientation (defined by the geometry of the emitter) is opposite, there will be no energy dissipation: the Poynting vector will curl into a ring, closing in on itself (Fig. 4b). As a result, the energy supplied to the supertor will begin to accumulate and gain the ability to exist autonomously, apart from the emitter it no longer needs. That is, in the form of an electromagnetic phantom, like an invisible ball lightning.

What I saw in the basement and thought was some kind of trickery could really be happening and was a demonstration of an amazing physical effect!

As the saying goes, never say "never," ...

Fields, molecules, stars

Perhaps the most surprising thing about this story was that I could neither see the author at the time of the demonstration, nor could I find him afterwards. And I could not find a single publication on the subject.

They may object to me speaking - saying that the work was classified. Damn it! The only things that are classified are those for which money is given, and money is only given for things that at least remotely resemble the truth. Anyway, what does a ghost look like, even if it is electromagnetic? In any case, I have often told physicists, who think quite open-mindedly, about this effect.

And guess what? My story did not even meet with polite interest...

At one time I really wanted to reproduce and study the electromagnetic phantom effect myself. However, this idea quickly had to be abandoned precisely because of the lack of money necessary for a serious study. All I did was to make several supertors, one of which you can see in the photo at the beginning of the article. Actually, this is not a super-, but a supersupertor, i.e., a spiral torus of the fourth order, or 1,2,3,4-torus.

For what purpose did I increase the order of the torus? The idea was clear enough. With a certain configuration of the emitter (most likely, left-left or right-right) two mutually perpendicular self-supported circular flows of electromagnetic energy (Fig. 4c) should arise in it within the structure of the physical vacuum element formed during annihilation of electron-positron pair – pseudoscalar $\tilde{Z} = \bar{E} \cdot \bar{H}$ (Fig. 5); you can read about this in detail in my article published in the “Russian Chemical Journal”, 1994, vol. 38, № 6, pp.107-117.

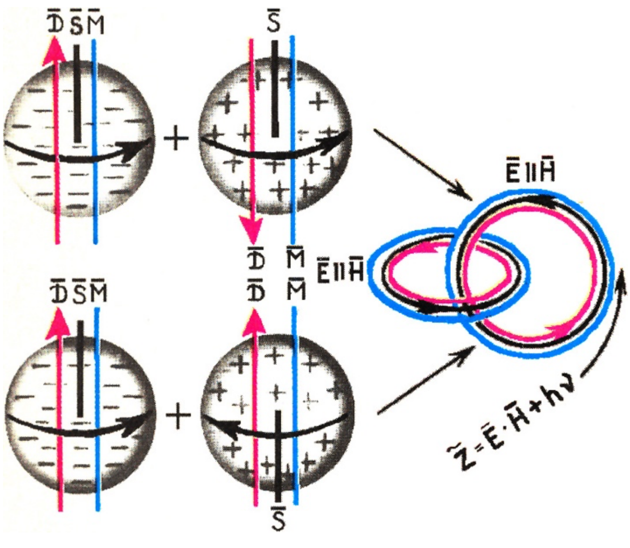


Fig. 5

As a result of the annihilation of the electron-positron pair, the rest mass turns into electromagnetic radiation; but since the electron and positron have spin $\bar{S}$ and magnetic and electric dipole moments $\bar{M}$ and $\bar{D}$, oriented in a certain way relative to each other, then where the annihilation occurred, there is also a chargeless electromagnetic field $\tilde{Z} = \bar{E} \cdot \bar{H}$, having the density of the energy flux.

In my opinion, such an electromagnetic phantom should have not only increased stability, but also the ability to draw from the external environment the very energy due to which, according to N.A. Kozyrev, stars shine (see “Chemistry of Life", 1994, No. 7, pp. 9-17) and which serves as the driving force of all processes of self-organization of matter.

By the way, the configuration of the electromagnetic fields that arise in a supersupertor is very similar to the configuration of the fields created to contain the plasma in stellarators - installations for the study of thermonuclear fusion.

And when I showed the supersupertor to a biologist I knew and asked him what it looked like, he exclaimed: “It's a model of a DNA molecule!”

That's how you play the bagel game ...